

BATTERY INPUT + PROTECTION

5V CONTROL BUCK (ESP32 + LOGIC ONLY)

WHEEL ENCODERS – BUILT INTO J4/J5 (3.3V)

BATTERY PI 5V / 5A BUCK – TPS54560B

SERVO 6V / 5A BUCK – TPS54560B

terminal, not 0.1in headers, for Pi power. Keep this 5A path short and wide.

reference design. Follow TI layout guidance; do not use this rail for motors or servos.

ESP32–PICO–KIT V4.1 SOCKETS

RASPBERRY PI UART / CONTROL

SAFETY / REVIEW NOTES

- VNH5019 exposed pads need reflow, thermal vias, and large copper pours.
- Verify actual motor stall current before ordering production PCB.
- Pi power is NOT made by U1. Feed J8 from a separate regulated 5V/5A supply.
- Never power ESP32 from USB and EXT_5V simultaneously.
- External fuse and latching E-stop are mandatory.

3-DOF ARM SERVO CONTROL

LEFT MOTOR DRIVER

RIGHT MOTOR DRIVER

Pi power uses dedicated TPS54560B 5 V / 5 A buck
DRAFT: verify motor stall current and footprints before manufacture
3S LiPo; 2x GA25–370; prototype assumption: 3 A peak per motor

Harrison

Sheet: /
File: pickleball_robot_controller.kicad_sch

Title: Pickleball Robot Wheel Controller + Raspberry Pi Interface

Size: A3 Date: 2026–07–12

Rev: R1–DRAFT

KiCad E.D.A. 10.0.4

Id: 1/1